DHVANI — Tech Stack

Adaptive software-defined sonar transmitter · SIH 2026, PS 26058 · breadboard prototype

Everything the current build runs on: hardware, firmware, tools, and the numbers that tie them together. The core idea in one line: a hardware timer paces DMA, DMA streams waveform samples from RAM into a GPIO port, an 8-bit DAC turns them into current, and op-amps turn that into a clean voltage — while the CPU sleeps. Water sensors pick the next chirp.

Signal chain

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Turbidity / TDS ──► ADC1 ──► firmware picks the chirp band
Firmware builds the chirp ──► waveform buffer in RAM
TIM1 update event ──► GPDMA1 ──► GPIOE->ODR  (PE0-PE7, one byte per sample)
DAC0800 (8-bit, current out) ──► TL074 A (current → voltage)
TL074 B (low-pass filter) ──► TL074 C (buffer) ──► scope (51 Ω) + buzzer (1 kΩ)
```

1 · HARDWARE

Layer	Component	Role	Key specs
Controller	**NUCLEO-U575ZI-Q** (STM32U575ZI, Arm Cortex-M33)	Runs firmware, timer, DMA, ADC	160 MHz, FPU, 786 KB RAM, SMPS, GPDMA
DAC	**DAC0800** (16-pin DIP)	Turns 8-bit samples into current	8-bit, ~100 ns settling, no clock pin, ~1 mA full scale here
Analog	**TL074** quad op-amp (14-pin DIP)	Amp A I→V · Amp B filter · Amp C buffer · Amp D spare	3 MHz GBW, 13 V/µs slew
Filter	Unity-gain Sallen-Key low-pass (1.5 kΩ, 150 pF)	Smooths DAC steps	Cut-off ≈ 700 kHz
Power	Nucleo +5 V and 3V3 · 9 V battery as −9 V rail	Dual supply for DAC0800 and TL074	+5 V / −9 V; 3V3 as DAC reference
Sensors	Turbidity module · TDS module (analog)	Water-condition inputs for adaptation	Read on ADC1 IN8 (PA3) and IN4 (PC3)
Output	Scope via 51 Ω · passive piezo buzzer via 1 kΩ	Ultrasonic proof · audible demo	Scope for >20 kHz, buzzer 2–17 kHz
Build	Solderless breadboard, solid-core wire	Proof-of-concept platform	Fine to ~5 MHz; KiCad PCB is the December step
Test gear	Multimeter · oscilloscope · INA226 (power, optional)	Gates, waveforms, energy per ping	

Pin map

Signal	U575 pin	Nucleo header	Goes to
DAC data B1 (MSB) … B8 (LSB)	PE7 … PE0	CN10/CN9, PE1 on CN11 pin 61	DAC0800 pins 5 … 12
Timer output / scope trigger	PA8 (TIM1_CH1)	CN10 pin 31	Scope trigger (not used by DAC0800)
Turbidity	PA3 (ADC1_IN8)	Zio A0, CN9 pin 1	Sensor via divider if >3.3 V
TDS	PC3 (ADC1_IN4)	Zio A2, CN9 pin 5	Sensor
+5 V / 3V3 / GND	—	CN8 pins 9 / 7 / 11	Rails, DAC reference
Status LED	PC7 (LD1)	On board	Toggles per ping / step

2 · FIRMWARE

Layer	What we use	Configuration
Language	C (C11), newlib, `math.h` with hardware FPU	sinf / cosf for waveform synthesis
Drivers	**STM32U5 HAL** + **BSP** (stm32u5xx_nucleo)	BSP for LEDs, button, ST-LINK serial (COM1, 115200)
Clock	MSIS 48 MHz → PLL1 (÷3 ×10 ÷1)	**160 MHz** SYSCLK, Voltage Scale 1, flash latency 4
Power	SMPS regulator	Low-power supply mode
Cache	ICACHE	1-way (direct mapped)
GPIO	Port E, PE0–PE11 push-pull outputs	Very High speed; PE0–PE7 drive the DAC
Timer	**TIM1**	Prescaler 0, Counter Period 159 → **1 MSps** sample clock; update event triggers DMA
DMA	**GPDMA1 Channel 0**, standard request mode	TIM1_UP, memory→peripheral, byte width, one transfer per ping
ADC	**ADC1**, 12-bit, scan of 2 channels	Offset calibration at start-up, long sampling time
Low power	`__WFI()` sleep during each ping	CPU idle while DMA streams
HAL time base	TIM17	Frees SysTick

Firmware modules

Function	Job
`sample_rate()`	Reads the timer so every waveform uses the true sample rate
`build_tone(f)`	Hann-windowed tone burst into the RAM buffer (8-bit, centre 128)
`build_chirp(f0, f1)`	Hann-windowed linear chirp — the sonar ping
`fire_ping()`	Starts one DMA transfer paced by TIM1; callback marks it done
`read_sensors()`	Reads turbidity and TDS on ADC1
Adaptation loop	Smooths turbidity, maps it to the chirp's top frequency, rebuilds, fires, sleeps, repeats

3 · TOOLS

Tool	Used for
STM32CubeMX (inside CubeIDE)	Pinout, clock tree, peripheral setup, code generation
STM32CubeIDE	Build, flash and debug over the on-board ST-LINK
ST-LINK (on the Nucleo)	Programming, debugging, virtual COM port
KiCad	PCB design — December
Reference documents	UM2861 (Nucleo-U575ZI-Q user manual), DAC0800 datasheet, TL074 datasheet

4 · THE NUMBERS

Parameter	Value
Resolution	8-bit (256 levels) → ~48 dB noise floor
Sample rate	1 MSps baseline; ceiling measured in step 9.3
Audible demo band	2–17 kHz chirp (`AUDIBLE 1`)
Ultrasonic band	100–250 kHz chirp at 1 MSps (`AUDIBLE 0`); more at 2 MSps+
Ping	20 000 samples = 20 ms at 1 MSps, then 200 ms quiet
Window	Hann — first sidelobe about −31 dB, above the 8-bit floor
Output swing	0 to ~3.0 V (1 mA × 3 kΩ feedback)
Adaptation	Muddier water → lower chirp band (high frequencies attenuate faster)

5 · WHAT WE DELIBERATELY DON'T USE

- **No FPGA.** Timer-paced DMA on the MCU delivers the sample stream with the CPU asleep.
- **No ML in the adaptation core.** A direct, explainable sensor → frequency mapping.
- **No circular DMA.** One transfer per ping suits sonar and avoids U5 linked-list setup.
- **No DS18B20 or pressure sensor** in the 24-hour build — they don't change the demo.

> **Honest limits to state on the slides:** 8-bit DAC; breadboard proof-of-concept; DAC0800 and TL074 chosen for availability, so the low-power case rests on the architecture (DMA, CPU asleep, duty-cycled pings) and the December PCB; no high-voltage transducer driver yet — the scope trace and buzzer are the proof.